Figure 1 assets: global-time slot labels

Predicted latent-state slots

$$\hat{z}^{1}_{t+1}, \quad \hat{z}^{1}_{t+2}, \quad \dots, \quad \hat{z}^{1}_{t+H}$$

Predicted action slots

$$\hat{a}^{1}_{t}, \quad \hat{a}^{1}_{t+1}, \quad \dots, \quad \hat{a}^{1}_{t+H-1}$$

Real environment-state slots

$$s_t, \quad s_{t+1}, \quad s_{t+2}, \quad \dots, \quad s_{t+H}$$

Encoded state-feedback slots

$$z_{t+1}, \quad z_{t+2}, \quad \dots, \quad z_{t+H}$$

Committed actions from the preceding chunk

$$a^{\mathrm{prev}}_{t}, \quad a^{\mathrm{prev}}_{t+1}, \quad \dots, \quad a^{\mathrm{prev}}_{t+H-1}$$

Aligned cross-chunk action overlap

$$i \in \mathcal{I}^{A}_{t}, \qquad a^{\mathrm{prev}}_{t+i}, \qquad i = 0, \dots, H-1$$

Figure 1 assets: stage-wise FBFM formulas

Execution transition and encoded feedback

$$
\begin{aligned}
s_{t+i} &\sim P\Big(\cdot \mid s_{t+i-1}, a^{\mathrm{prev}}_{t+i-1}\Big)\,, \\
z_{t+i} &= E(\mathcal{O}(s_{t+i}))\,, \qquad (i, z_{t+i}) \in \mathcal{F}_{t,k}
\end{aligned}
$$

State-flow correction

$$
\begin{aligned}
\mathbf{e}^{Z}_{t,k} &= P_Z \mathbf{W}^{Z}_{t,k}\left(\mathbf{Y}^{Z}_{t,k} - \hat{\mathbf{Z}}^{1}_{t,k}\right), \\
v^{Z}_{\mathrm{FBFM}} &= v^{Z}_{\theta_Z} + \lambda^{Z}_{\tau^{Z}_{k}}\left(\mathbf{J}^{Z}_{t,k}\right)^{\mathsf{T}} \mathbf{e}^{Z}_{t,k}
\end{aligned}
$$

Action-flow correction

$$
\begin{aligned}
\mathbf{e}^{A}_{t,k} &= P_A \mathbf{W}^{A}_{t}\left(\mathbf{Y}^{A}_{t} - \hat{\mathbf{A}}^{1}_{t,k}\right), \\
v^{A}_{\mathrm{FBFM}} &= v^{A}_{\theta_A} + \lambda^{A}_{\tau^{A}_{k}}\left(\mathbf{J}^{A}_{t,k}\right)^{\mathsf{T}} \mathbf{e}^{A}_{t,k}
\end{aligned}
$$

Feedback-refreshed state context

$$
\begin{aligned}
\check{\mathbf{Z}}_{t,k} &= \left(\mathbf{I}_{D_Z} - \mathbf{W}^{Z}_{t,k}\right)\hat{\mathbf{Z}}_{t} + \mathbf{W}^{Z}_{t,k}\mathbf{Y}^{Z}_{t,k}, \\
\mathcal{C}^{Z}_{t,k} &= \Phi_Z\big(\check{\mathbf{Z}}_{t,k}, \mathcal{H}_t\big)
\end{aligned}
$$

Figure 3 assets: joint-generation FBFM

Joint state–action variable

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{Z}_t \\ \mathbf{A}_t \end{bmatrix}, \qquad v_\theta^X : \mathbf{X}_t^\tau \mapsto \frac{\mathrm{d}\mathbf{X}_t^\tau}{\mathrm{d}\tau}$$

Joint feedback target, mask, and preconditioner

$$\mathbf{Y}_{t,k}^X = \begin{bmatrix} \mathbf{Y}_{t,k}^Z \\ \mathbf{Y}_t^A \end{bmatrix}, \qquad \mathbf{W}_{t,k}^X = \begin{bmatrix} \mathbf{W}_{t,k}^Z & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_t^A \end{bmatrix},$$
$$\mathbf{P}^X = \mathrm{blkdiag}\big(P_Z \mathbf{I}_{D_Z}, P_A \mathbf{I}_{D_A}\big)$$

Joint guided velocity field

$$\mathbf{e}_{t,k}^X = \mathbf{P}^X \mathbf{W}_{t,k}^X \left(\mathbf{Y}_{t,k}^X - \hat{\mathbf{X}}_{t,k}^1 \right),$$
$$v_{\mathrm{FBFM}}^X = v_\theta^X + \lambda_{\tau_k^X}^X \left(\mathbf{J}_{t,k}^X \right)^\mathsf{T} \mathbf{e}_{t,k}^X$$

Direct state-to-action correction

$$\mathbf{e}_{t,k}^A = \mathbf{0} \quad \Longrightarrow \quad \mathbf{g}_{t,k}^A = \mathbf{J}_{ZA}^\mathsf{T} \mathbf{e}_{t,k}^Z = \left(\frac{\partial \hat{\mathbf{Z}}_{t,k}^1}{\partial \mathbf{A}_t^{\tau_k^X}} \right)^\mathsf{T} \mathbf{e}_{t,k}^Z$$

Shared replacement formulas and compact labels

Generic FBFM flow integral

$$\mathbf{Q}_t^1 = \mathbf{Q}_t^0 + \int_0^1 v_{\mathrm{FBFM}}^Q(\mathbf{Q}_t^\tau, \tau)\,\mathrm{d}\tau, \qquad Q \in \{Z, A, X\}$$

Unified clean-endpoint and guidance chain

$$\begin{aligned}
\hat{\mathbf{Q}}_{t,k}^1 &= \mathbf{Q}_t^{\tau_k^Q} + (1-\tau_k^Q) v_{\theta_Q}^Q\left(\mathbf{Q}_t^{\tau_k^Q}, \tau_k^Q; \mathcal{K}_{t,k}^Q\right), \\
\mathbf{e}_{t,k}^Q &= \mathbf{P}^Q \mathbf{W}_{t,k}^Q\left(\mathbf{Y}_{t,k}^Q - \hat{\mathbf{Q}}_{t,k}^1\right), \\
v_{\mathrm{FBFM}}^Q &= v_{\theta_Q}^Q + \lambda_{\tau_k^Q}^Q\left(\mathbf{J}_{t,k}^Q\right)^{\mathsf{T}} \mathbf{e}_{t,k}^Q
\end{aligned}$$

Measurement and generalized lifting

$$\begin{aligned}
\mathbf{m}_{t,k}^Q &= h_Q(\mathbf{Q}_t^1) + \boldsymbol{\eta}, \\
\mathbf{Y}_{t,k}^Q &:= h_Q^\dagger(\mathbf{m}_{t,k}^Q), \qquad Q \in \{Z, A, X\}
\end{aligned}$$

Stage-wise factorization

$$p_\theta(\mathbf{Z}_t, \mathbf{A}_t \mid \mathcal{H}_t) = p_{\theta_Z}(\mathbf{Z}_t \mid \mathcal{H}_t)\, p_{\theta_A}(\mathbf{A}_t \mid \mathbf{Z}_t, \mathcal{H}_t)$$

Compact diagram labels

Compact text labels

State Flow	v^{Z}_{FBFM}
Action Flow	v^{A}_{FBFM}
Joint State–Action Flow	v^{X}_{FBFM}
Dynamic State Feedback	$\mathcal{F}_{t,k}$
Committed Action Overlap	$\mathcal{I}^{A}_{t}$
Frozen WAM	$\theta, \theta_Z, \theta_A$ fixed